



IC 240: Mechanics of Rigid Bodies

Course Code	IC 240
Course Name	Mechanics of Rigid Bodies
L-T-P-C	1.5-1.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; 44th BoA (Nov 2021); 37th Senate (Oct 2022), in the B.Tech Institute Core (IC) baskets - all branches curriculum
Revised	44th BoA (Nov 2021) - revised

Course Contents

- **Equilibrium:** System isolation and the free body diagram, equilibrium conditions [7 Lecture]
- **Structures:** Introduction, plane trusses, method of joints and method of sections, frames and machines. [7 Lecture]
- **Applications of friction** [6 Lecture]
- **Kinematics of Rigid Bodies:** Introduction, rotation, absolute motion, relative velocity, instantaneous center of zero velocity, relative acceleration, motion relative to rotating axes. [10 Lecture]
- **Kinetics of Rigid Bodies:** Introduction, general equations of motion, translation, fixed axis rotation, general plane motion, Work-energy relations, virtual work, Impulse momentum equations. [12 Lecture]

Text Books

1. J. L. Meriam, L.G. Kraige; **Engineering Mechanics: Statics**; Willey India Pvt. Ltd.
2. J. L. Meriam, L.G. Kraige; **Engineering Mechanics: Dynamics**; Willey India Pvt. Ltd.

References

1. Beer, Johnston, Eisenberg, Sarubbi; **Vector Mechanics for Engineers Statics and Dynamics**, McGraw Hill Company
2. S. P. Timoshenko, D.H. Young; **Engineering Mechanics**, McGraw-Hill Book